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Redox Reactions

NEET Chemistry · Previous Year Questions with Answer Key

34	26	1988 – 2026	English
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MCQ (Single Correct Answer)	34 questions
Answer key	at the end

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MCQ (Single Correct Answer) 34 questions

Q1. RE-NEET 2026

In an acidic medium, 10 mL of 0.25 M oxalic acid is titrated with $KMnO_4$ solution. If the volume of $KMnO_4$ solution required to reach end point is 10 mL , the strength of the $KMnO_4$ solution is

(A) 0.15 M (B) 0.10 M (C) 0.20 M (D) 0.25 M

Open solution

Q2. RE-NEET 2026

The correct decreasing order of oxidation state of the $\underline{\text{underlined atom}}$ in each molecule is

(A) $\underline{P}_4O_6 > \underline{Cl}_2O_7 > \underline{Al}H_3$

(B) $\underline{P}_4O_{10} > \underline{S}O_3 > H_2\underline{O}$

(C) $\underline{N}_2O_5 > \underline{Al}_2O_3 > H_2\,\underline{S}$

(D) $\underline{Pb}O_2 > \underline{N}_2O_3 > \underline{S}O_3$

Open solution

Q3. NEET 2025

Consider the following compounds :

$\underline{K}O_2, H_2\underline{O}_2$ and $H_2\underline{S}O_4$

The oxidation state of the underlined elements in them are, respectively,

(A) $+1, -2$, and +4 (B) $+4, -4$, and +6 (C) $+1, -1$, and +6 (D) $+2, -2$, and +6

Open solution

Q4. NEET 2024 (Re-Examination)

The oxidation states not shown by Mn in given reaction are :

$3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O$

A. +6

B. +2

C. +4

D. +7

E. +3

Choose the most appropriate answer from the options given below :

(A) D and E only (B) B and D only (C) A and B only (D) B and E only

Open solution

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Q5. NEET 2024

Which reaction is NOT a redox reaction?

(A) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

(B) $2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$

(C) $H_2 + Cl_2 \rightarrow 2HCl$

(D) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

Open solution

Q6. NEET 2023 Manipur

The correct option for a redox couple is:

(A) Both are oxidised forms involving same element.

(B) Both are reduced forms involving same element.

(C) Both the reduced and oxidised forms involve same element.

(D) Cathode and anode together.

Open solution

Q7. NEET 2023

On balancing the given redox reaction,

$aCr_2O_7^{2-} + bSO_3^{2-}(aq) + cH^+(aq) \rightarrow 2aCr^{3+}(aq) + bSO_4^{2-}(aq) + \frac{c}{2}H_2O(\ell)$

the coefficients a, b and c are found to be, respectively -

(A) 3, 8, 1 (B) 1, 8, 3 (C) 8, 1, 3 (D) 1, 3, 8

Open solution

Q8. NEET 2022 Phase 2

Which of the following reactions is a decomposition redox reaction?

(A) $P_4(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$

(B) $2Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$

(C) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

(D) $Cl_2(g) + 2OH^-(aq) \rightarrow ClO^-(aq) + Cl^-(aq) + 4H_2O(l)$

Open solution

Q9. NEET 2020 Phase 2

The oxidation number of the underlined atom in the following species

Identify the incorrect option.

(A) $H\underline{Au}Cl_4$ is $+3$

(B) $\underline{Cu_2O}$ is -1

(C) $\underline{ClO_3^-}$ is $+5$

(D) $K_2\underline{Cr_2}O_7$ is $+6$

Open solution

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Q10. NEET 2020 Phase 1

What is the change in oxidation number of carbon in the following reaction?
$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(l) + 4HCl(g)$

(A) 0 to + 4 (B) -4 to + 4 (C) 0 to $-$ 4 (D) + 4 to + 4

Open solution

Q11. NEET 2019 (Odisha)

The oxidation state of Cr in CrO_6 is

(A) $+4$ (B) -6 (C) $+12$ (D) $+6$

Open solution

Q12. NEET 2019

Which of the following reactions are disproportionation reaction?

(a) $2Cu^{+} \rightarrow Cu^{2+} + Cu^{0}$
(b) $3MnO_4^{2-} + 4H^{+} \rightarrow 2MnO_4^{-} + MnO_2 + 2H_2O$
(c) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$
(d) $2MnO_4^{-} + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^{\oplus}$

Select the **correct** option from the following :

(A) (a), (c) and (d) (B) (a) and (b) only (C) (a) and (d) only (D) (a), (b) and (c)

Open solution

Q13. NEET 2018

The correct order of N-compounds in its decreasing order of oxidation states is

(A) HNO_3 , NO, N_2 , NH_4Cl (B) HNO_3 , NO, NH_4Cl, N_2
(C) HNO_3 , NH_4Cl, NO, N (D) NH_4Cl, N_2 , NO, HNO_3

Open solution

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Q14. NEET 2018

For the redox reaction,

$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$

The correct coefficients of the reactants for the balanced equation are

Open solution

(A)

MnO_4^-	$C_2O_4^{2-}$	H^+
16	5	2

(B)

MnO_4^-	$C_2O_4^{2-}$	H^+
2	5	16

(C)

MnO_4^-	$C_2O_4^{2-}$	H^+
2	16	5

(D)

MnO_4^-	$C_2O_4^{2-}$	H^+
5	16	2

Q15. NEET 2016 Phase 2

Hot concentrated sulphuric acid is a moderately strong oxidizing agent. Which of the following reactions does not show oxidizing behaviour?

Open solution

(A) $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

(B) $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

(C) $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$

(D) $CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$

Q16. AIPMT 2014

The pair of compounds that can exist together is

Open solution

(A) $FeCl_3$, $SnCl_2$ (B) $HgCl_2$, $SnCl_2$ (C) $FeCl_2$, $SnCl_2$ (D) $FeCl_3$, KI

Q17. AIPMT 2014

(I) $H_2O_2 + O_3 \rightarrow H_2O + 2O_2$

(II) $H_2O_2 + Ag_2O \rightarrow 2Ag + H_2O + O_2$

Role of Hydrogen peroxide in the above reactions is respectively

Open solution

(A) oxidizing in (I) and reducing in (II)

(B) reducing in (I) and oxidizing in (II)

(C) reducing in (I) and (II)

(D) oxidizing in (I) and (II)

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Q18. AIPMT 2014

In acidic medium, H_2O_2 changes $Cr_2O_7^{2-}$ to CrO_5 which has two

$(-O-O-)$ bonds. Oxidation state of Cr in CrO_5 is

(A) +5 (B) +3 (C) +6 (D) -10

Open solution

Q19. AIPMT 2012 Prelims

When Cl_2 gas reacts with hot and concentrated sodium hydroxide solution, the oxidation number of chlorine changes from

(A) zero to +1 and zero to -5 (B) zero to -1 and zero to +5

(C) zero to -1 and zero to +3 (D) zero to +1 and zero to -3

Open solution

Q20. AIPMT 2012 Prelims

A mixture of potassium chlorate, oxalic acid and sulphuric acid is heated. During the reaction which element undergoes maximum change in the oxidation number?

(A) S (B) H (C) Cl (D) C

Open solution

Q21. AIPMT 2009

Oxidation numbers of P in PO_4^{3-}, of S in SO_4^{2-} and that of Cr in $Cr_2O_7^{2-}$ are respectively

(A) +3, +6 and +5 (B) +5, +3 and +6 (C) -3, +6 and + 6 (D) +5, +6 and +6

Open solution

Q22. AIPMT 2008

Number of moles of MnO_4^- required to oxidize one mole of ferrous oxalate completely in acidic medium will be

(A) 7.5 moles (B) 0.2 moles (C) 0.6 moles (D) 0.4 moles

Open solution

Q23. AIPMT 2007

The number of moles of $KMnO_4$ that will be needed to react with one mole of sulphite ion in acidic solution is

(A) $\frac{4}{5}$ (B) $\frac{2}{5}$ (C) 1 (D) $\frac{3}{5}$

Open solution

Q24. AIPMT 2004

Which is the best description of the behaviour of bromine in the reaction given below?

$H_2O + Br_2 \rightarrow HOBr + HBr$

(A) Proton acceptor only (B) Both oxidised and reduced

(C) Oxidised only (D) Reduced only

Open solution

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Q25. AIPMT 2003

The oxidation states of sulphur in the anions

SO_3^{2-}, $S_2O_4^{2-}$ and $S_2O_6^{2-}$ follow the order

(A) $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$

(B) $SO_3^{2-} < S_2O_4^{2-} < S_2O_6^{2-}$

(C) $S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}$

(D) $S_2O_6^{2-} < S_2O_4^{2-} < SO_3^{2-}$

Open solution

Q26. AIPMT 1999

Oxidation state of Fe in Fe_3O_4 is

(A) $\frac{3}{2}$ (B) $\frac{4}{5}$ (C) $\frac{5}{4}$ (D) $\frac{8}{3}$

Open solution

Q27. AIPMT 1999

In which of the following compounds transition metal has zero oxidation state?

(A) $[Fe(CO)_5]$ (B) $NH_2 \cdot NH_2$ (C) $NOClO_4$ (D) CrO_5

Open solution

Q28. AIPMT 1997

Which of the following is redox reaction?

(A) Evaporation of H_2O

(B) Both oxidation and reduction

(C) H_2SO_4 with NaOH

(D) In atmosphere O_3 from O_2 by lighting.

Open solution

Q29. AIPMT 1996

Reaction of sodium thiosulphate with iodine gives

(A) tetrathionate ion

(B) sulphide ion

(C) sulphate ion

(D) sulphite ion.

Open solution

Q30. AIPMT 1996

$KMnO_4$ reacts with oxalic acid according to the equation

$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

Here 20 mL of 0.1 M $KMnO_4$ is equivalent to

(A) 50 mL of 0.5 M $C_2H_2O_4$

(B) 20 mL of 0.1 M $C_2H_2O_4$

(C) 20 mL of 0.5 M $C_2H_2O_4$

(D) 50 mL of 0.1 M $C_2H_2O_4$

Open solution

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Q31. AIPMT 1995

The oxide, which cannot act as a reducing agent is

(A) CO_2 (B) ClO_2 (C) NO_2 (D) SO_2

Open solution

Q32. AIPMT 1994

Which substance is serving as a reducing agent in the following reaction?

$14H^+ + Cr_2O_7^{2-} + 3Ni \rightarrow 7H_2O + 2Cr^{3+} + 3Ni^{2+}$

(A) H^+ (B) $Cr_2O_7^{2-}$ (C) H_2O (D) Ni

Open solution

Q33. AIPMT 1994

The oxidation state of I in $H_4IO_6^-$ is

(A) $+1$ (B) -1 (C) $+7$ (D) $+5$

Open solution

Q34. AIPMT 1988

The oxidation state of Cr in $K_2Cr_2O_7$ is

(A) +5 (B) +3 (C) +6 (D) +7

Open solution

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Answer Key

1. **B**	2. **C**	3. **C**	4. **D**	5. **D**
6. **C**	7. **D**	8. **B**	9. **B**	10. **B**
11. **D**	12. **B**	13. **A**	14. **B**	15. **D**
16. **C**	17. **C**	18. **C**	19. **B**	20. **C**
21. **D**	22. **C**	23. **B**	24. **B**	25. **A**
26. **D**	27. **A**	28. **B**	29. **A**	30. **D**
31. **A**	32. **D**	33. **C**	34. **C**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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